MIL-H-3177C
6 Febraurary1976
SUPERSEDING
MIL-H-317713
1 September 1966

# MILITARY SPECIFICATION

# HEATERS, COOLANT, ENGINE:

# MUTIFUEL-BURNING, 24-VOLT D.C .

This specification is approved for use by all Departments and Agencies of the Department of Defense.

## 1. SCOPE

1.1 Scope. This specification covers 24-volt direct-current, multifuel-burning, forced-draft, engine-coolant heaters.

1.2 Classification. The coolant heaters shall be of the following nominal capacities, as specified (see 6.2):

Capacity A - 15,000 Btu per hour.
Capacity B - 30,000 Btu per hour.
Capacity C -60,000 Btu per hour.

## 2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

SPEC IFICATIONS

Federal

| | |
|---|---|
| VV-F-800 | - Fuel Oil, Diesel. |
| VV-G-76 | - Gasoline, Automotive. |
| PPP-B-601 | - Boxes, Wood, Cleated-Plywood. |
| PPP-B-621 | - Boxes, Wood, Nailed and Lock-Corner. |
| PPP-B-636 | - ‘Boxes, Shipping, Fiberboard. |
| PPP-F-320 | - Fiberboard; Corrugated and Solid, Sheet ‘Stock (Container Grade ), and cut shapes l |
| PPP-P-40 | - Packaging and Packing of Hand Tools. |

FSC 2990

Source: https://assist.dla.mil -- Downloaded: 2026t0-16T17:15Z
Check the source to verify that this is the currentversion before use.

Military

| | |
|---|---|
| MIL-P-116 | - Preservation-Packaging Methods of. |
| MIL-V-173 | - Varnish, Mositure-and-Fungus-Resistant (for Treatment of Communications, Electronic, and Associated Equipment ). |
| MIL-P-514 | - Plates, Identification, Instruction and Marking, Blank. |
| MIL-T-704 | - Treatment and Painting of Materiel. |
| MIL-G-3056 | - Gasoline, Automotive, Combat. |
| MIL-C-5015 | - Connectors Electric, "AN" Type l |
| MIL-T-5624 | - Turbine Fuel, Aviation, Grades JP-4 and JP-5. |
| MIL-L-6085 | - Lubricating oil: Aircraft Instrument, Low volatility. |
| MXL-W-6858 | - Welding, Resistance: Aluminum, Magnesium, Nonhardening Steels or Alloys, Nickel Alloys, Heat-Resisting Alloys, and Titanium Alloys; Spot and Seam. |
| MIL-W-8611 | - Welding, Metal Arc and Gas, Steels, and Corrosion and Heat Resistant Alloys; Process for. |
| MIL-A-1175 | - Antifreeze, Arctic-Type. |
| MIL-C-13486 | - Cable, Special Purpose, Electrical: Low-Tension, Heavy Duty Single Conductor and Multiconductor. |
| MIL-P-14105 | - Paint, Heat-Resisting (For Steel Surfaces) . |
| MIL-G-23827 | - Grease, Aircraft and Instrument, Gear and Actuator Screw. |
| MIL-L-46167 | - Lubricating Oil, Internal-Combustion Engine, Arctic. |

STANDARDS

Military

| | |
|---|---|
| MIL-STD-105 | - Sampling Procedures and Tables for Inspection by Attributes. |
| MI-STD-129 | - Marking for Shipment and Storage. |
| MIL-STD-130 | - Identification Marking of US Military Property. |
| MIL-STD-461 | - Electromagnetic Interference Characteristics Requirements for Equipment l |



Source: https://assist.dla.mil -- Downloaded: 2026-10-16T17:15Z
Check the source to verify that this is the current version before use.

| | |
|---|---|
| MIL-STD-810 | - Environmental Test Methods. |
| MIL-STD-889 | - Dissimilar Metals. |
| MS25331-4 | - Light Assembly, Press to Test Indicator. |
| MS39061 | - Switch, Toggle, SPST, SPDT, 24 Volt, D.C., 25 Amp (Waterproof). |
| MS51321 | - Pump, Fuel, Electrical -24 Volts D.C., 25 G.P.H. Capacity. |

(copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN IRON AND STEEL INSTITUTE (AISI)

Steel Products Manual

(Application for copies should be addressed to the American Iron and Steel Institute, 150 East 42nd Street, New York, NY 10017. )

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

Power Test Code - Supplement on Instruments and Apparatus.
Part 10 - Flue must Gas Analysis.

(Application for copies should be addressed to the American Society of Mechanical Engineers, 345 East 47th Street, New York, NY 10017 . )

COUNTY OF LOS ANGELES - AIR POLLUTION CONTROL DISTRICT

Rule 66 - Organic Solvents.

(Appication for copies should be addressed to the County of Los Angeles - Air Pollution Control District, 434 S. San Pedro Street, Los Angeles, CA 90003.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T17:15Z
Check the source to verify that this is the current version before use.

## 3. REQUIREMENTS

3.1 Description. The heater shall consist of a burner, combination combustion-chamber and heat-exchanger, combution-air blower and fuel-metering and evaporizing arrangement all assembled in one unit with necessary Integral control devices, plus compatible but separate accessories including a control box, wiring harness, fuel filter, fuel pump, coolant pump, and pressure relief valve. The heater shall be adaptable for installation in either a horizontal or vertical position within the overall dimensions shown in TabLe I and with the coolant discharge connection at the highest point.

Table I. Maximum Installation Dimensions

| Capacity | Position | Height | Width | Length |
|---|---|---|---|---|
| A | Horizontal | 11" | 7" | 17" |
| A | Vertical | 17 | 7 | 11 |
| B | Horizontal | 11 | 7 | 20 |
| B | Vertical | 20 | 7 | 11 |
| C | Horizontal | 13 | 9 | 26 |
| C | Vertical | 26 | 9 | 13 |

NOTE: The control box, fuel pump, fuel filter, coolant pump, or combustion air inlet damper (see 3.12) need not be confined within the dimensions specified in Table I. Full performance in tilted positions is also required (see 3.8.3.1).

3.2 First article (preproduction model). The supplier shall furnish two heaters for examination and testing within the time frame specified (see 6.2), to prove prior to starting production that his production methods will produce heaters that comply with the requirements of this specification. Examination and tests shall be as specified in Section 4; and, unless otherwise specified herein, all examination and tests shall be conducted by the supplier subject to surveillance and approval by the Government (see 6.3).



Source: https://assist.dla.mil -- Downloaded: 2026-10-16T17:15Z
Check the source to verify that this is the current version before use.

3.3 Material. Material shall be as specified herein. Materials not specified shall be selected by the supplier and shall be subject to all provisions of this specification.

3.3.1 Dissimilar metals. Nonpermissable galvanic couples (dissimilar metals) as defined in MIL-STD-889, Type II shall be insulated as specified in Section 5 of MIL-STD-889.

3.4 Fasteners. All screws, bolts, nuts, and other fasteners which are not inherently corrosion-resistant shall be cadmium-, zinc or nickel-plated to resist atmospheric corrosion under the environmental conditions specified herein. Sheet metal or self-tapping screws shall not be used.

305 Environmental conditions. The heater shall function under the following environmental conditions:

(a) During operation in ambient temperatures from plus 80" F to minus 65° F.
(b) After exposure to conditions of rain, humidity, salt fog, sand and dust, vibration, and shock as specified in MIL-STD-810 and in Section 4 herein.
(c) After storage at ambient temperatures of plus 155' F with dewpoint not over 85° F for 4 hours daily for a period of 2 days and at minus 65° F for 72 hours.

3.6 Safety. All moving parts and parts which are subject to high operating temperatures or are energized electrically shall be insulated, fully enclosed, or guarded. Protective devices shall not impair the operating functions. Fire hazard shall be minimized by automatic functions as specified in 3.8.3 and 3.8.4.2. Hazards associated with the handling or operation of the heater shall be identified as early as possible in the life cycle and eliminated or suitably controlled as required by MIL-STD-882. Possible sources of fuel leakage should be located where fuel cannot. contact. hot surfaces or electrical components.

3.7 Maintainability. The heater sahall provide for quick disconnection and remounting so that maintenance and adjustment can normally be accomplished indoors. When demounted., all parts and assemblies of the heater shall be accessible for servicing, repair, and adjustment. Service adjustments required shall be the minimum consistent with required performance. Insofar as practicable, adjustments shall not. be disturbed by disassembly required for cleaning and checking, or shall be readily returnable to correct settings. All practicable provision shall be made to prevent the possibility of disassembly of components and subassemblies.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T17:15Z
Check the source to verify that this is the current version before use.

3.8 Performance.

3.8.1 Heat output. When tested in accordance with 4.5.4.5, the heater shall conform to the heat output requirements specified in Table II.

Table II. Minimum Heat Output

| Capacity | Heat to Coolant Btu/hr | | Exhaust Back Pressure |
|---|---|---|---|
| A | 13,500 | at | .5" $H_2O$ |
| B | 27,000 | at | .75 $H_2O$ |
| C | 54,000 | at | 1.00 $H_2O$ |

The above minimum shall be otained when operating in minus 50° F ambient temperature, consuming diesel fuel in accordance with VV-F-800, Grade DF-A.

3.8.2 Fuel. The heater shall operate without adjustment and as specified herein on any of the following fuels:

(a) Gasoline conforming to MIL-G-3056, Type II.
(b) Jet fuel conforming to MIL-T-5624, Grades JP-4 and JP-5, in ambient temperatures above the pour and cloud points of the fuel.
(c) Gasoline conforming to VV-G-76, regular grade.
(d) Leaded or aromatic fuels up to 100-octane grade.
(e) Diesel fuel conforming to VV-F-800, Grades DF-A and DF-1, in ambient temperatures above the pour and cloud points of the fuel.

3.8.3 Cold starting. The heater shall start and operate at 19 volts electric input within the time periods specified using the fuels at the ambient temperatures specified in Table 111. Combustion shall become selfsustaining, clean, and steady within the specified time periods.

Table III. Cold Starting

| Ambient Temperature | Fuel | Time Period, Minutes |
|---|---|---|
| Minus 65° F | Gasoline, MIL-G-3056, Type II | 3 |
| Minus 65° F | Jet fuel, MIL-T-5624, Grade JP-4 | 3 |
| Minus 50° F | Diesel fuel, VV-F-800, Grade DF-A | 4 |

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T17:15Z
Check the source to verify that this is the current version before use.

When specified (see 6.2), an automatic control system shall carry out the designed starting cycle after the main control switch is operated. The automatic control system shall cut off electric power to the heater fuel pump, and coolant pump when combustion is not obtained within 4 minutes after the control switch is turned to "ON" position. Manual reset of this stitch or of a circuit breaker shall be necessary to init iate another starting attempt after automatic cutoff because of combust ion failure.

3.8.3.1 Cold starting and operation in tilted positions. The heater shall start and operate as specified herein continuously for not less than 1 hour when tilted 15 degrees in each of the four directions from level position in either the horizontal or vertical Installation-adaptation (see Table 1).

3.8.4 Operation. After combustion is stabilized with each of the fuels at the temperature specified in Table III, operating characteristics of the heaters shall be as specified in Table IV. The power input values shall be obtained with 24-volt dc power supply and shall include inputs to the fuel pump and coolant pump With coolant conforming to MIL-A-11755 initially at ambient temperature at the pump intake. Power input requirements in Table IV are to be measured after heater ignition and & not include ignition requirements. Exhaust temperatures under Table IV are to be measured 12 inches beyond point of discharge from heater.

Table IV. Operating Characteristics

| Capacity | Power Input, Watts, Total Max. | Fuel Rate, Lb/Hr Max. | Exhaust Tem, Max. |
|---|---|---|---|
| A | 150 | 1.8 | 1,200° F |
| B | 250 | 3.6 | 1,200° F |
| C | 400 | 7.2 | 1,200° F |

3.8.4.1 Combustion. During stabilized operation in accordance with 3.8.4 above (including each of the three temperature-fuel combinations), combustion shall continue to be self-sustaining, clean, and steady. Carbon dioxide content of' the exhaust gas shall be not less than 8.0 percent, and the smoke reading shall be not more than 1 on a Bacharach smokemeter or the equivalent reading on other approved device.



Source: https://assist.dla.mil -- Downloaded: 2026-10-16T17:15Z
Check the source to verify that this is the current version before use.

3.8.4.2 Automatic functioning. The following functions In addition to those required for starting (see 3.8.3) shall. be accomplished automatically:

(a) Purge unburned fuel and cool combustion chamber after every initiation of the "cutoff" cycle regardless of whether initiation is accomplished manually or automatically. The blower and coolant pump shall continue to operate during this purge period. The time period for purging shall not exceed 4 minutes.
(b) Cutoff operation in event of combustion failure resulting from exhaustion-of fuel supply or blockage of inlet air. Manual reset shall then be necessary to reinitiate the starting cycle.
(c) Cutoff operation when the temperature of coolant returning to the heater reaches 100° to 120° F, and start it again when the coolant-return temperature reaches 50° to 60° F.
(d) Keep an indicator lamp lighted on the control box or panel, whenever combustion is sustained in the heater.
(e) Cutoff operation if the coolant temperature should reach 230" F at the outlet of the heater. This cutoff shall require manual reset to restore operation.
(f) Protect motors against overheat and provide one or more circuit breakers as practicable to protect circuitry against shorts or overload.

3.8.5 Service life. When tested as specified in 4.5.4.12, the heater sahll operate without structural failure. There shall be not more than two occurrences of leakage or of malfunction or performance-deterioration requiring cleaning, adjustment, or replacement of parts.

3.8.6 Electromagnetic interference, The electromagnetic interference emission and susceptability characteristics of the heaters shall conform to MIL-STD-461, Class III C.

3.9 Fuel pump. The electrical fuel pump furnished with each heater shall conform to] MS51321-2. The pump shall be controlled by the control system of the heater.

3.10 coolant pump. The coolant pump may be integral with or separate from the heater. In either case the pump shall be controlled by the neater control system, stall operate whenever there is sustained combustion



Source: https://assist.dla.mil -- Downloaded: 2026-10-16T17:15Z
Check the source to verify that this is the current version before use.

in the heater, and shall discharge into the heater jacket. Pump inlet and outlet (except if the pump is integral) shall be threaded, 3/4 inch NPT, female, and fitted with hose-to-pipe adapter for 3/4 inch ID hose. The pump shall have a permanent shaft seal and shall be integral with and driven by a totally-enclosed rotor, fungus treated in accordance with 3.18. Motor bearings shall be of a type not requiring lubrication or shall be the lubricated-for-life type. When tested in minus 65" F ambient temperature with intake head of 20 inches of coolant (see 4.5.4.10), the pumps for the three capacities of heater shall be as specified in Table V.

Table V. Coolant Pump Characteristics

| Heater Capacity | Discharge Pressure, Minimum psig | Flow, Minimum gpm |
|---|---|---|
| A | 1.75 | 0.25 |
| B | 3.50 | 0.50 |
| C | 7.00 | 1.00 |

Note: The pump, if integral with the heater, shall be as specified in Table V with the intake head reduced to 6 inches of coolant. Flow readings to be taken at minus 65" F, 5 minutes after ignition or 5 minutes after heater indicator light actuates when heater and pump are installed in a closed circuit system of 5 gallon capacity.

3.11 Coolantt connections. Quick-disconnect couplings shall be provided at the coolant inlet and outlet of the heater. Unless otherwise] specified (see 6.2), the couplings shall be 3/4 inch IPS external type with self-sealing valves, equal to and Interchangeable with model VEN12 as -- ufactured by Snap-Tite, Incorporated, Union City, PA. A male half-coupling shall be installed on the coolant outlet and a female half-coupling on the inlet. The mating half-couplings for these shall be furnished and shall have adapters for 3/4-inch ID hose. The coolant outlet shall also provide a pressure relief valve adjustable from 10 to 40 psig opening-pressure and a vent to bleed trapped air from the heater jacket.

3.12 Combustion air inlet. The combustion air inlet shall be circular and shall provide within the dimensions specified in Table I a 3/4-inch-long neck for attachment of a duct, but shall provide a snap-on screened cover for this opening. When specified (see 6.2), an electrically-actuated poppet valve or damper shall also be furnished which will provide a dust-tight closure when the heater is not operating but will be open wherever the blower is running. This damper need not be confined within the overall dimensions prescribed in Table I.



Source: https://assist.dla.mil -- Downloaded: 2026-10-16T17:15Z
Check the source to verify that this is the current version before use.

3.13 Combustion air blower. The combustion air blower shall supply the required amount of air to insure complete fuel combustion and limit exhaust temperature in accordance with Table IV (see 3.8.1). The blower shall be driven by an integrally-mounted electric motor fully enclosed and fungus treated in accordance with 3.18. Motor bearings shall be of a type not requiring lubrication or shall be of the lubricated-for-life type.

3.14 Combustion chamber and coolant jacket. The heater combustion chamber and coolant jacket or heat exchanger shall be made of an Austenitic, 18-8 stainless steel of the AISI 30C series which is consistent with the life, corrosion resistance, and other design considerations of the assembly. The coolant jacket or heat exchanger shall withstand a pressure of 35 psig (see 4.5.4.11).

3.15 Control box. The control box shall be suitable for installation at a point accessible to the operator and shall have the following components Installed:

(a) Main control switch conforming to MS39061 (see 3.8.3).
(b) Main automatic circuit breaker. Maybe combined with (a) (see 3.8.4.2(f)).
(c) Conbbustion-Indicator lamp (see 3.8.4 .2(d)) (MS25331-4).
(d) Test button, lamp and connections to check for availability of power at the heater. May be connected to use same lamp as (c).

3.16 Electrical harness. Electrical cable shall conform to MIL-C-13486. Quick-disconnect multiconductor cable terminals and receptacles shall conform to MIL-C-5015. Quick-disconnects shall be provided for both ends of the cable from the control box to the heater and for the heater end of the cables to the coolant pump and fuel pump. The control-box-to-heater cable assembly shall be not less than 10 feet in length overall.

3.17 Fungus and moisture resistance. The electrical circuitry, including all components and connections except as specified below, shall be protected from the effects of fungus growth and moisture by an overall treatment with a varnish conforming to MIL-V-173, composition as specified in 3.18.1, with 1 percent copper 8-quinolinolate (by weight) based on the nonvolatile content of the varnish:

(a) Components or circuit elements that are inherently fungus and moisture resistant or which are hermetically sealed need not be treated.



Source: https://assist.dla.mil -- Downloaded: 202640-16T17:15Z
Check the source to verify that this is the current version before use.

(b) Components or circuit elements whose functions will be adversely affected by the varnish coating shall not be treated.

When used, the varnish shall be applied by spray, brush, or a combination of both to give a dry-film thickness of not less than 1 mil to component or element surfaces previously cleaned and prepared so that the surfaces are free from all foreign matter which would interfere with the adherence or function of the varnish.

3.17.1 Composition. Composition I shall be used unless Composition. II must be used in order to comply with local air pollution regulations in the application of the varnish. When Composition II is used, the supplier shall provide evidence to the Government that the use of Composition II is required, and shall certify that the Composition II material complies with Rule 66 - Air Pollution Control District, County of Los Angeles, CA.

3.18 Delivery lubrication service. When delivered, all necessary lubrication shall be installed. Grease lubrication, including lubrication of sealed bearings, shall be with grease conforming to MIL-G-23827, as applicable, and oil lubrication with oil conforming to MIL-L-46167 or MI\L-L-6085 .

3.19 Identification marking. The heater shall be identified in accordance with MIL-STD-130. Identification plates shall conform to MIL-P-514, Type 1, Style 2, Composition C, of Type 1, Grade A, Class 1 material.

3.19.1 Instruction plates. An instruction plate (see 3.20) containing a schematic and a wiring diagram shall be furnished for the heater. The wiring diamgram shall portray the physic- location of all tiring inter connections in the same relationship as when they are installed. Identification markings of wires, instruments, control devices, and connections shall be as shown in both diagrams. Instruction plates or diagrams shall also be furnished, ir.eluding warnings and cautions, describing My special or important procedures to be followed in assembling, operating, or servicing the heater.

3.20 Treatment and painting. The portions of the heater and its components and parts normally painted shall be cleaned, treated, and painted in accordance with MIL-T-704, Type A, color as specified (See 6.2). Parts subject to heat shall be coated as necessary with paint conforming to MIL-P-14105.

3.21 Workmanship.

3.2.1 Metal fabrication. Metal used in fabrication shall be free from kinks and sharp bends. The straighter of material shall be done by methods that will not cause injury to the material. Corners shall be



Source: https://assist.dla.mil -- Downloaded: 2026-10-16T17:15Z
Check the source to verify that this is the current version before use.

square and true. Flame cutting, using tips suitable for the thickness of the steel may be employed instead of shearing and sawing. All bends shall be made with controlled means to insure uniformity of size and shape. Precaution shall be taken to avoid overheating. Heated steel shall be allowed to Cool slowly. External surfaces shall be free of burrs, sharp edges and corners, except when sharp edges or corners are required or where they are not detrimental to safety.

3.21.2 Bolted connections. Boltholes shall be accurately formed and shall have burrs removed. Washers or lockwashers shall be provided where necessary. Matching thread areas securing bolts conforming to SAE J429 or capscrews shall be of sufficient strength to withstand the tensile strength of the bolt. All fasteners shall be correctly torqued and shall have full thread engagement. Bolts shall protrude not more than two full threads.

3.21.3 Riveted connections. Rivets shall fill the holes completely. The upset rivet heads shall be full, neatly made, concentric with the rivet holes, and in full contact with the surface of the member, and shall be in accordance with SAE J492.

3.21.4 Welding. All welding shall be done in accordance with MIL-W-8611 or MIL-W-6858.

3.22 Noise. The steady-state noise level shall not exceed Category D, Table II of MIL-sTD-1474.

3.23 Human factors. The characteristics shall provide for operation by personnel in all type clothing and shall be designed in compliance with MIL-STD-1472.

3.24 Vibration. The heater shall be designed to withstand the vibration tests as described in 4.5.4.9.2 without structural damage, malfunction of components, or any other irregular operation.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government The Government reserves the right to perform any of the inspections set set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to presribed requirements.



Source: https://assist.dla.mil -- Downloaded: 2026-10-16T17:15Z
Check the source to verify that this is the current version before use.

4.2 Classification of inspections. Inspections shall be classified as follows: :

(a) Preproduction inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).
(c) Inspection of preparation for delivery (see 4.6).

4.3 Preproduction inspection.

4.3.1 Examination. The heaters shall be examined as specified in Table VI. Presence of one or more defects shall be cause for rejection.

4.3.2 Tests. Two preproduction model heaters shall be tested as specified in Table VI. One shall be tested In the vertical position, the other in the horizontal (see Table I). Tests shall be conducted in the sequence listed, except where otherwise specified. Failure of any test shall be cause for rejection.

4.4 Quality conformance inspection.

4.4.1 Sampling. Sampling for examination and test shall be in accordance with MIL-STD-105, Inspection Level S-4 for test.

4.4.2 Examination. Samples-selected in accordance with 4.4.1 shall be examined as specified In Table VI. AQL shall be 2.5 percent defective for major defects and 6.5 percent defective for minor defects.

4.4.3 Tests. Samples selected in accordance with 4.4.1 shall be tested as specified in Table VI. AQL shall be 2.5 percent defective. Approximately one-third Of the heaters to be tested in each lot shall be tested in the vertical-installation adaptation, the remaining items in the horizontal adaptation. Tests shall be conducted in the sequence listed, except as otherwise specified.

4.5 Inspection procedure.

4.5.1 Inspection schedule. The inspection schedule is contained in Table VI.

Table VI. Inspection Schedule

| Sequence | Preproduction | Quality Conformance | Test | Inspection Paragraph | Requirement Paragraph |
|---|---|---|---|---|---|
| 1 | X | X | Examination. | 4.5.2 | - |
| 2 | X | X | Test conditions. | 4.5.3 | - |
| 3 | X | X | Test measurement. | 4.5.3.1 | - |
| 4 | X | - | Starting, -65° F, gasoline. | 4.5.4.1 | 3.8.3 |



Source: https://assist.dla.mil -- Downloaded: 2026-10-16T17:15Z
Check the source to verify that this is the current version before use.

Table VI. Inspection Schedule (Cont'd)

| Sequence | Preproduction | Quality Conformance | Test | Inspection Paragraph | Requirement Paragraph |
|---|---|---|---|---|---|
| 5 | X | - | Start-failure-control, -65° F, gasoline. | 4.5.4.2 | 3.8.3 |
| 6 | X | - | Tilted start and operation, -65° F, gasoline. | 4.5.4.3 | 3.8.3.1 |
| 7 | X | - | Operating characteristics, -65° F, gasoline. | 4.5.4.4 | 3.8.4, 3.8.4.1 |
| 8 | X | - | Automatic functioning tests, -65° F, gasoline. | 4.5.4.6 | 3.8.4.2 |
| 9 | X | - | Starting, -65° F, jet fuel. | 4.5.4.1 | 3.8.3 |
| 10 | X | - | Tilted start and operation, -65° F, jet fuel. | 4.5.4.3 | 3.8.3.1 |
| 11 | X | - | Operating characteristics, -65° F, jet fuel. | 4.5.4.4 | 3.8.4 3.8.4.1 |
| 12 | X | X | Starting, -50° F, diesel fuel. | 4.5.4.1 | 3.8.3 |
| 13 | - | X | Start-failure-control, -50° F, diesel fuel. | 4.5.4.2 | 3.8.3 |
| 14 | X | X | Tilted start and operation, -50° F, diesel fuel. | 4.5.4.3 | 3.8.3.1 |
| 15 | X | X | Operating characteristics and heat output, -50° F, diesel fuel. | 4.5.4.4 | 3.8.1, 3.8.4, 3.8.4.1 |
| 16 | - | X | Automatic functioning tests, -50° F, diesel fuel. | 4.5.4.6 | 3.8.4.2 |
| 17 | X | - | Environmental tests. | 4.5.4.9* | 3.5 |



Source: https://assist.dla.mil -- Downloaded: 2026-10-16T17:15Z
Check the source to verify that this is the current version before use.

Table VI. Inspection Schedule (Control)

| Sequence | Preproduction | Quality Conformance | Test | Inspection Paragraph | Requirement Paragraph |
|---|---|---|---|---|---|
| 18 | X | - | Starting, -50° F, diesel fuel. | 4.5.4.1 | 3.8.3 |
| 19 | X | - | Start-failure-control, -50° F, diesel fuel. | 4.5.4.2 | 3.8.3 |
| 20 | X | - | Operating characteristics and heat output, -50° F, diesel fuel. | 4.5.4.4, 4.5.4.5 | 3.8.1, 3.8.4, 3.8.4.1 |
| 21 | X | - | Automatic functioning tests, -50° F, diesel fuel. | 4.5.4.6 | 3.8.4.2 |
| 22 | X | - | Service life. | 4.5.4.12 | 3.8.5 |
| 23 | X | - | Start-failure-control, -50° F, diesel fuel. | 4.5.4.2 | 3.8.3 |
| 24 | X | - | Operating characteristics and heat output, -50° F, diesel fuel. | 4.5.4.4, 4.5.4.5 | 3.8.1, 3.8.4, 3.8.4.1 |
| 25 | X | - | Automatic functioning tests, -50° F, diesel fuel. | 4.5.4.6 | 3.8.4.2 |

Note: The following tests of components may be made at any time without regard to sequence or to coordinating with any of tests tabulated above.

| | | | | |
|---|---|---|---|---|
| X | - | Motor-overheat. | 4.5.4.7 | 3.8.4.2(f) |
| X | - | Circuit-breaker. | 4.5.4.8 | 3.8.4.2(f) |
| X | - | Coolant pump. | 4.5.4.10 | 3.10 |
| X | X | Coolant pressure. | 4.5.4.11 | 3.14 |
| X | - | Electromagnetic interference. | 4.5.4.13 | 3.8.6 |

* Including provisions of its subparagraphs.



Source: https://assist.dla.mil -- Downloaded: 2026-10-16T17:15Z
Check the source to verify that this is the current version before use.

4.5.2 Examination. The heaters shall be examined as specified herein for the following defects:

Major

101. Dimensions not in accordance with Table I.
102. Voltage not as specified.
103. Components missing, not as specified, damaged, or defective.
104. Leakage of fuel, coolant, or exhaust.
105. Assembly incomplete or incorrect.
106. Identification or special marking missing, illegible, or incorrect.

Minor

201. Fastenings missing or not tightened.
202. Workmanship not as specified.
203. Treatment and painting not as specified.

4.5.3 Test conditions. Tests shall be conducted utilizing and connecting all components, such as control box, fuel pump and filter, coolant pump with connecting cables, and hoses and quick-disconnects, as Specified or as appropriate, except as tests of separate components are specified. Fuel supply electric power, coolant beat-sink and exhaust connections shall be provided, together with controls and facilities as necessary for accurate regulation of test conditions. No repair, realinement, or adjustment shall be performed on quality conformance test units after their selection or after start of test nor on preproduction test units after start of test except as specified in 3.8.5. Components subject to atmospheric corrosion shall not be treated with any treatment such as grease or oil in order to meet test requirements. Any lubrication required shall be accomplished with military lubricants and prior to testing. Test fuels shall be those specified in Table III; certified analyses shall be submitted by the supplier to prove conformity with specifications.

4.5.3.1 Test measurement. Measurements of test factors shall be as follows :

(a) Electric input. A dc voltmeter shall be connected to the heater terminals and a dc ammeter shall be connected into the circuit, with external shunt if required. Calibrated accuracy of the voltmeter and of t-he ammeter with external shunt, if used, shall be within 5 percent of the values measured.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T17:15Z
Check the source to verify that this is the current version before use.

(b) Fuel input. Fuel shall be measured directly by weight or by volume. Measured volume shall be multiplied by known specific weight of fuel. Overall accuracy shall be 2 percent or better.

(c) Temperature.

(1) Exhaust gas. Temperature shall be measured by thermocouple, preferably chromel-alumel, in center of exhaust-outlet pipe and not more than 12 inches from the combustion chamber. Accuracy shall be within 10° F, in the range from 500" to 1500° F.

(2) Ambient temperature shall be measured by two or more radiation-shielded thermocouples preferably copper-constantan, or by radiation-shielded resistance elements. Accuracy shall be within 2° F.

(3) Coolant temperature at the heater inlet and outlet shall be measured by an immersion thermocouple with junction in a 1/8-inch-diameter protecting tube, installed with the junction in the center of the coolant line and as close as practicable to the heater- Accuracy shall be within 2° F, absolute and temperature-differential. between inlet and outlet of the heater shall be determinable within 0.259 F.

(d) Exhaust back-pressure shall be measured by a manometer tapped into the exhaust line and close as practicable to the heater. The manometer shall be carefully zeroed, and precaution shall be taken to avoid clogging the connecting tube with frost during cold tests. Overall accuracy shall be within 5 percent.

(e) Carbon-dioxide and smoke in exhaust. The carbon-dioxide content of the exhaust gas shall be measured with an Orsat apparatus as prescribed in the ASME Power Test Code, Supplement on Instruments and Apparatus, Part 10, Flue Exhaust Gas Analyses, or equal, as approved by the contracting officer. The smoke measurement shall be in accordance with 3.8.4.1 unless an alternate device is approved by the contracting officer.

(f') Coolant pressures shall be measured by mercury-manometer or calibrated pressure gage of suitable scale-

(g) Coolant flow may be measured by timed flow-diversion to a measuring container, using a stopwatch if diversion times are leas than 1 minute; alternatively by a flowmeter calibrated under the same conditions and flow ranges occurring in these tests l



Source: https://assist.dla.mil -- Downloaded: 2026-10-16T17:17Z
Check the source to verify that this is the current version before use.

4.5.4 Tests.

4.5.4.1 Starting. Soak-out the test chamber, heater, fuel, and apparatus thoroughly to test temperature before each trial. Check the exhaust line to make sure that any water or unburned fuel will drain away from the heater without accumulating at any point. Set the exhaust back-pressure control device at the same setting used to create back pressures specified in 3.8.1 when operating under the same conditions. Control the electric voltage at not more than 19 volts under starting load. operate the starting control and measure the time to obtain self-sustaining, clean, and steady combustion. Nonconformance to the time limits specified in Table III, back firing, or obvious malfunctioning shall constitute failure of this test.

4.5.4.2 Start-failure-control. Drain the fuel from the heater and disconnect it from the fuel supply. Soak-out the test-chamber, heater, and apparatus to test temperature. Set the power supply to normal (approximately 26 V) voltage. Operate the starting control and measure the time to automatic cutoff. Reconnect the fuel supply, soak-out for at least 1 hour, set up special precautions against possible fire or explosion. Reduce the supply voltage to the minimum which will operate the fuel pump when the starting control is turned on. Reset, operate the starting control, and measure the time to automatic cutoff or to start. " If cutoff occurs, bring supply voltage back to normal, wait 1 minute, then reset, operate the starting control again, and measure the time to start. Failure to cut off within the time specified in 3.8•3 in the no-fuel trial, failure to cut off or start within the same specified time In the low-voltage trial., or failure to start within the time limit specified in Table III with normal voltage after cutoff in the low-voltage trial, backfiring, or obvious malfunctioning shall constitute failure of this test.

4.5.4.3 Tilted start and operation. With the heater in each of the tilt positions specified in 3.8.3.1, test in accordance with 4. 5. 4.1 except continue operation for 1 hour after each start. Failure to meet the time limits given in Table III, backfiring or obvious malfunctioning, or failure to continue operation for 1 hour after each start shall constitute failure of this test.

4.5.4.4 operating characteristics. Soak-out the test chamber, heater, fuel, and the apparatus In the chamber to specified temperature. Start the heater, control-the supply voltage and the coolant temperature at pump intake to values specified in 3.8.4 and the exhaust back pressure specified in 3.8.1 with stabilized heater operation. Measure and record total electric power input, fuel rate, exhaust temperature, carbon dioxide) and smoke. Nonconformance to 3.8.4 and 3.8.4.1 shall constitute failure of this test.



Source: https://assist.dla.mil -- Downloaded: 2026-10-16T17:15Z
Check the source to verify that this is the current version before use.

4.5.4.5 Heat output. In ambient temperature and on fuel specified in 3.8.1, apply procedure or 4.5.4.4 except add measurement and recording of coolant flow and temperatures of coolant at inlet and outlet of heater. Nonconformance to 3.8.1, 3.8.4, and 3.8.4.1 sahll constitute failure of this test.

4.5.4.5.1 Calculation of heat output. Determine output of the heater by the following formula:

$$Q = WC (T_2 - T_1)$$

where:

Q = measured output, Btu/hr.
W = flow rate of antifreeze solution, lb/hr.
C = specific heat of antifreeze solution.
$T_1$ = temperature of antifreeze solution entering heater, °F.
$T_2$ = temperature of antifreeze solution leaving heaters °F.

4.5.4.6 Automatic functioning. Determine conformance to 3.8.4.2 by tests in the low ambient temperatures and on the fuels specified in 4.5.1, and in accordance with the following procedures.

4.5.4.6.1 Purge and cut off after exhaustion of fuel supply. Start the heater and operate until stable combustion is attained. Cut off fuel supply by valving-off or by disconnecting power from fuel pump. Inab ility to purge and cut off all other functions within 4 minutes shall constitute failure of this test.

4.5.4.6.2 Cutoff air inlet-air blockage. Reset, start heater and operate until stable combustion is attained. Block air inlet. Inability of the blower and coolant pump to continue to run but to cut off within 4 minutes shall constitute failure of this test.

4.5.4.6.3 Cycling control. With heater in normal operation, control temperature of coolant entering the heater so that it rises gradually until automatic cutoff and purge occurs. Permit the heater and its coolant to cool down until it automatically restarts. Observe several cycles and record. Inability to Start or stop automatically and consistently Within the temperature specified in 3.8.4.2(c) shall constitute failure of this test.



Source: https://assist.dla.mil -- Downloaded: 2026-10-16T17:15Z
Check the source to verify that this is the current version before use.

4.5.4.6.4 Overheat control. Short-circuit the cycling-control thermostat or otherwise prevent it from cutting off heater operation. With the heater in normal operation, control the cod-ant-outlet temperature 80 that it rises slowly until automatic cutoff and purge occurs. Reduce the temperature of the coolant at least 50° F below the overheat cutoff point (see 3.8.4.2(e)). When separate coolant pump is used disconnect the coolant pump from its power supply and valve-off coolant circulation by means of a valve near the heater inlet. On integral pump units restrict coolant flow of inlet by valving or other means to check cutoff point against design cutoff temperature. Reset control and start the heater, operate until automatic cutoff occurs again. On the first trial with slow temperature buildup, inability to initiate cutoff within 5° F of the temperature specified in 3.8.4.2(e) sahll constitute failure of the test. On the second trial with blocked coolant circulation, temperature criteria shall not apply, but inability to cut off end purge without coolant blowoff, hammering, or other evidence of excessive coolant temperature shall constitute failure of that test.

4.5.4.7 Motor overheat. This test shall be applied to the pump and blower motors. Place the motor in a small test cell with electric leads. connected and brought out so that a circuit continuity check can be made. Raise the temperature in the test cell slowly while circuit continuity is checked frequently. Record the temperature at which the overheat control opens the circuit and compare with the motor manufactures design data - Also disassemble and inspect the motor for evidence of damage to the windings and other components. A difference of more than 10°- F from the designer's cutoff point or damage to the motor shall constitute failure of this test.

4.5.4.8 Circuit breaker.

4.5.4.8.1 Starting current. Rig the blower motor and coolant pump motor to block their rotors from turning. Connect the heater to a power source which can maintain at least 22 volts under the current draw



Source: https://assist.dla.mil -- Downloaded: 2026-10-16T17:15Z
Check the source to verify that this is the current version before use.

when the heater control is turned on. Failure of the breaker to function before there is any evidence of overheating of the motors or any damage to the motors or other components, or indication of internal damage upon disassembly after test, shall constitute failure of this test.

4.5.4.8.2 Short circuit. Disconnect the circuit breaker from other components and subject it to a current several times as high as normal starting current, Failure of the breaker to function immediately shall constitute failure of this test.

4.5.4.9 Environmental. The test setup shall be as follows: Cover the air inlet if the cover is hinged or "captive" and supplied as part of the heater. Fit to the exhaust outlet an unsupported section of metal hose or tube not more than 15 inches long, with its outlet turned downward. Install quick-disconnect fuel, electric, and coolant fittings, and relief and vent valves. Make connections to the pump and to a surge tank with at least 12 inches of unsupported 3/4-inch ID hose on the pump inlet and the heater outlet. Fill this coolant system with antifreeze solution. Fill fuel system with diesel fuel. Make up electrical connections complete including leads to source of power. Operation during environmental tests shall not be required but after being subjected to each environment the major assemblies may be wiped dry, cleaned with rags, brush, air-jet, water spray detergent-solution insofar as can be done in 10 minutes by one man without any disassembly requiring tools. Immediately thereafter inspect for damage, misalinement, and leakage, then conduct check test by starting the heater, operating, and stopping, to check for any basic malfunction. If minor changes are made to enable the heater to pass any one test which would be likely to reduce capability on other previously conducted tests, the affected tests or all previous tests shall be repeated, at the discretion or the contracting officer.

4.5.4.9.1 Rain, humidity, salt-fog, sand, and dust. Conduct the rain, humidity, salt-fog, and sand and dust tests in accordance with MIL-STD-810, Methods 506, 507, 509, and 510, respectively, except that limited cleanup after test shall be permitted in accordance with 4.5.4.9 above before each start operation check. Malfunction on any check test shall constitute failure of the test.

4.5.4.9.2 Vibration. Conduct the vibration test in accordance with MIL-STD-810, Method 514, equipment Class 5 (ground vehicles), Curve B of Figure 514-5 except that maximum frequency shall be 400 cps, minimum 7 cps and except as otherwise specified herein. The complete heater setup as described in 4.5.4.9 shall be mounted together with the control box; however, the surge tank need not be on the vibration table. Brackets, straps, bolts, and shock mounts used for installation and furnished as



Source: https://assist.dla.mil -- Downloaded: 2026-10-16T17:15Z
Check the source to verify that this is the current version before use.

part of the heater shall be used for the test installation; no other braces, straps, tiedowns, or auxiliary devices shall be used, either externally or internally. There s.nail be no resonance in the test fixture between 7 and 400 cps. Removable covers and access doors shall be in place during all testing except during the resonance search which will be made during tne initial part of the sinusoidal-cycling-run. Separate accelerometers shall be installed for each main support of the heater and for the pump and the control box. In event of differences in the "G" values measured by these accelerometers, the vibration acceleration shall be controlled to keep the least of the corrected readings at or above 2.5G. Structural damage, misalinement, leaks of fuel or coolant, or component-malfuction shall constitute failure of these tests.

4.5.4.9.2.1 Resonance search. Conduct the resonance search using vibration-acceleration input less than 2.5G If supplier so elects; however, if this is done the time shall not be credited toward fulfillment of the sinusoidal-cycling requirements. Open up the heater as much as practicable to observe resonance of internal parts. Use stroboscope, high-intensity lights, or similar devices to facilitate the detection and evaluation of the resonances.

4.5.4.9.3 Shock. Conduct the shock tests in accordance with MIL-STD-810, Method 516, Procedure I. Structural damage, misalinement, leak of fuel or coolant, or component malfunction shall constitute failure of these tests.

4.5.4.9.4 High-temperature storage . Subject the heater to an ambient temperature of 155° F for a period of 4 hours per day for 2 days. structural damage, misalinement, leak of fuel or coolant, or component malfunction shall constitute failure of these tests.

4.5.4.9.5 low-temperature storage. Subject the heater to an ambient temperature of minus 65° F for a period of 72 hours. Structural damage, misalinement, leak of fuel or coolant, or component malfunction shall constitute failure of these tests. (Note: This test may be scheduled separately from others in the environmental series, may use the same facilities used for the cold start and heat output repeat tests and in coordination with those tests.)

4.5.4.10 Coolant pump. Pump antifreeze conforming to MIL-A-11755 and control input-voltage to the motor to 24 volts. Nonconformance to Table V shall constitute failure of this test.

4.5.4.11 Coolant pressure. Fill the heater with water or antifreeze, connect to pressure source, and apply liquid pressure of at least 35 pounds . Maintain this pressure for at least 5 minutes. Evidence of leaks or permanent distortion shall constitute failure of this test.

Source: https://assist.dla.mil -- Downloaded: 2025-10-07T17:15Z
Check the source to verify that this is the current version before use.

$.5.4.12 Service life. Subject the preproduction model heaters (one in vertical, one in horizontal installation-adaptation) to a service-life-simulation series of preheat cycles according to the following procedure and the schedule in Table VII. In each cycle, cool, and soak-out the heater to the Specified ambient temperature, than start and operate with gradually increasing coolant temperatures until cutoff by the cycling control occurs, followed by normal purging. Then control coolant temperature to drop gradually until restart by the] cycling control occurs. Continue to operate until the prescribed time requirement for that cycle is fulfilled. (Coolant temperature may be controlled to prevent more than one automatic cutoff and restart per preheat cycle. Cooldown time shall be at least 5 minutes from automatic cutoff to restart. Soak-out time before each cycle shall. be at least equal to the specified operating time for that cycle, after thermocouples Indicate full cooldown of heater and coolant.) Complete the prescribed number of cycles on each fuel in the sequence gasoline, jet fuel, and diesel fuel, before starting any cycles at the next lower temperature. Nonconformance to 3.8.5 shall constitute failure] of this test.

Table VII. Service Lift Test

| Number of Cycles on Each Fuel | | | Ambient Temperature Degrees F | Minimum Operating Time Per Cycle Minutes |
|---|---|---|---|---|
| Gasoline | JP-4 | Diesel | | |
| 50* | 50* | 50* | Normal | 10 |
| 20 | 20 | 20 | 0 | 20 |
| 15 | 15 | 15 | -20 | 30 |
| 10 | 10 | 10 | -40 | 40 |
| 5 | 5 | 5 | -50 | 60 |

*Note: Arctic grade fuels will not be required for the tests in normal temperatures.

4.5.4.13 Electromagnetic interference. The preproduction model shall be tested to determine conformance to 3.8.6. The supplier shall furnish the contracting officer a report of tests required by MIL-STD-461. Disapproval of the report shall constitute failure of this test.

4.5.4,14 Noise level. Noise levels shall be determined in accordance with the procedures contained in MIL-STD-1474. Nonconformance to 3.23 shall constitute failure of this test.

4.5.4.15 Human factors. The heater shall be evaluated throughout the testing to determine that the human factors criteria of 3.24 have been met.



Source: https://assist.dla.mil -- Downloaded: 2026-10-16T17:15Z
Check the source to verify that this is the current version before use.

4.5.4.16 Maintenance evaluation. Maintenance operations required during testing shall be accomplished as specified in 3.70 Any assemblies or support equipement such as technical manuals, repair parts, special tools, etc., provided under the contract for operation or conducting maintenance will be evaluated.

4.6 Inspection of preparation for delivery.

4.6.1 Quality conformance inspection of pack.

4.6.1.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.6.1.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.6.1.3 Examination. Samples selected in accordance with 4.6.1.2 shall be examined for the following defects. AQL shall be 2.5 percent defective.

| | |
|---|---|
| 107 l | Materials, methods, and containers not as specified. Each incorrect material, method, and container shall be considered one defect. |
| 108. | Contents not blocked, braced, and cushioned as required for Level A or B. |
| log , | Strapping not as specified for Level A. |
| 110. l | Boxes not lined when required to prevent abrasion as specified for Level A or B. |
| 111. | Marking illegible, incorrect, or incomplete. |
| 112 l | Fiberboard boxes not closed as specified for Level B. |
| 113. | Commercial packaging not in accordance with referenced documents. |

5. PREPARATION FOR DELIVERY

5.1 Preservation. Preservation shall be Level A, Level B, or Commercial as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Maintenance tools. Maintenance tools as required shall be preserved in accordance with PPP-P-40, Level A.



Source: https://assist.dla.mil -- Downloaded: 2026-10-16T17:15Z
Check the source to verify that this is the current version before use.

5.1.1.2 Heater. The heater and its accessories including repair parts and technical publications shall be preserved Method IA-15 of MIL-P-116. The inner container shall conform to PPP-E-636, Class Domestic, grade as required to meet the size and weight limitations of the box specification. The preserved maintenance tools, shall be placed in the box with the heater. The contents shall be blocked, braced and cushioned as required to prevent movement and damage.

5.1.2 Level B.

5.1.2.1 Maintenance tools. Maintenance tools shall be preserved as specified in 5.1.1.1 for Level A.

5.1.2.2 Heater. Each complete heater with repair parts and technical publication shall be preserved in accordance with Method I of MIL-P-116, and placed in a box along with the preserved maintenance tools, conforming to PPP-B-636, Class Weather Resistant, grade as required to meet the size and weight limitations of the box specification. The box shall be closed in accordance with Method V of the appendix to the box specification.

5.1.3 Commercial. Commercial preservation shall be in accordance with FED. No. 356.

5.2 Packing. Packing shall be Level A, Level B, or Commercial as specified (see 6.2).

5.2.1 Level A. Heaters, preserved and packaged as specified in 5.1, shall be packed in close-fitting boxes conforming to PPP-B-601, Overseas Type, Grade B, style optional, or PPP-B-621, Class 2, style optional. The boxes shall be closed and strapped in accordance with the appendix to the applicable box specification. Boxes for heaters, preserved and packaged as specified in 5.1.1 for Level A, shall be lined with fiberboard conforming to PPP-F-320, Class Weather Resistant, to prevent abrasion of the barrier material.

5.2.2 Level B. Heaters preserved and packaged as specified in 5.1 shall be packed as specified in 5.2.1 for Level A except the boxes shall be Domestic Type; or the boxes shall be in accordance with PPP-B-636, V3c. Fiberboard boxes shall be closed Method V of the box specification appendix.

5.2.3 Commercial. Commercial packing of the heaters and equipment shall be in accordance with FED. STD. No. 356.



Source: https://assist.dla.mil -- Downloaded: 2026-10-16T17:15Z
Check the source to verify that this is the current version before use.

5.3 Marking. Packages and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 intended use. The heaters are intended to be used primarily in facilitating the start of liquid-cooled, internal-combustion engines at ambient temperatures as low as minus 65° F. They also intended for standby heating but not for operation while the engine is running.

6.2 Ordering data. Procurement documents should specify the following:

(d Title, number, and date of this specification.
(b) Capacity required (see 1.2).
(c) Time frame required for submission of preproduction model (see 3.2).
(d) When automatic control is required (see 3.8.3).
(e) Coolant connections if other than as specified (see 3.32).
(f) Whether electrically operated, dusttight air-inlet closure is required (see 3.12).
(g) Color to be painted (see 3.21).
(h) Time interval for submission of electromagnetic interference reports (see 4.5.4.13).
(i) Preservation and packing required (see 5.1 and 5.2). Level B preservation, and packaging is intended to provide economical but limited protection and should be specified only when it is determined the heaters will be held in covered storage.

6.3 Preproduction model. Any changes or deviations of production heaters from the approved preproduction model during production will be subject to the approval of the contracting officer. Approval of the pre-production model will not relieve the supplier of his obligation to furnish heaters conforming to this specified (see 3.2).

6.4 Data requirements. The contracting officer should include requirements for such data as technical publications, instructional materials, illustrated parts lists, and supplier's maintenance and operation manual to be furnished with each heater.



Source: https://assist.dla.mil -- Downloaded: 2026-10-16T17:15Z
Check the source to verify that this is the current version before use.

6.5 provisioning. The contracting officer should include provisioning requirements for repair parts and maintenance tools as necessary (including any special tools), and instructions on shipment of heaters.

Custodians:

Army - ME
Air Force - 82

Review activities:

Army - AT, EL
DSA - CS

User activities:

Army - MI, AV
Navy - MC

Prapring activity:

Army - ME

Project No. 2990-0073

*US GOVERNMENT PRINTING OFFICE 1976- 603-766: 2993



Source: https://assist.dla.mil -- Downloaded: 2026-10-16T17:15Z
Check the source to verify that this is the current version before use.

# STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

*(See Instructions – Reverse Side)*

| 1. DOCUMENT NUMBER | 2. DOCUMENT TITLE |
|---|---|
| | |

| 3a. NAME OF SUBMITTING ORGANIZATION | 4. TYPE OF ORGANIZATION *(Mark one)* |
|---|---|
| | ☐ VENDOR<br>☐ USER<br>☐ MANUFACTURER<br>☐ OTHER *(Specify)*: _____ |
| b. ADDRESS *(Street, City, State, ZIP Code)* | |
| | |

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

| 7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional | b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional |
|---|---|
| c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional | 8. DATE OF SUBMISSION *(YYMMDD)* |
| | |

*(TO DETACH THIS FORM, CUT ALONG THIS LINE.)*

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

Source: https://assist.dla.mil -- Downloaded: 202610-16T17:15Z
Check the source to verify that this is the current version before use.

**INSTRUCTIONS:** In a continuing effort to make our standardization documents better, the DoD provides this form for use in submitting comments and suggestions for improvements. All users of military standardization documents are invited to provide suggestions. This form may be detached, folded along the lines indicated, taped along the loose edge *(DO NOT STAPLE)*, and mailed. In block 5, be as specific as possible about particular problem areas such as wording which required interpretation, was too rigid, restrictive, loose, ambiguous, or was incompatible, and give proposed wording changes which would alleviate the problems. Enter in block 6 any remarks not related to a specific paragraph of the document. If block 7 is filled out, an acknowledgement will be mailed to you within 30 days to let you know that your comments were received and are being considered.

*NOTE:* This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

*(Fold along this line)*

*(Fold along this line)*

DEPARTMENT OF THE ARMY

NO POSTAGE NECESSARY IF MAILED IN THE UNITED STATES

**OFFICIAL BUSINESS**
PENALTY FOR PRIVATE USE $300

**BUSINESS REPLY MAIL**
FIRST CLASS PERMIT NO. 12062 WASHINGTON D C.

POSTAGE WILL BE PAID BY THE DEPARTMENT OF THE ARMY

Commander
US Army Mobility Equipment Research & Development Command
ATTN: DRDME-DS
Fort Belvoir, VA 22060

Source: https://assist.dla.mil -- Downloaded: 202640-16T17:15Z
Check the source to verify that this is the current version before use.